

UNIVERSITY OF CLAUDE · COURSE SYLLABUS

CSCE 679 — Data Visualization

The encoding decides what can be read, so the encoding is the design

AT A GLANCE

Credits: 3 credit-equivalents · **Modules:** 13 · **Term:** Semester 11 (with CSCE 671 and CSCE 632)

Prerequisites: CSCE 671 Human-Computer Interaction, especially its Module 02 on perception; CSCE 676 Data Mining for the material being visualised; CSCE 647 Advanced Rendering for Module 12

Effort: 12–14 hours per week · 13 modules + 2 project weeks

Deliverable: A visualisation of a dataset you did not choose for its convenience — with the encoding justified against the perceptual ranking, the uncertainty shown, a baseline chart it has to beat, and readers tested on what they can actually extract from it

Course description

A visualisation is a mapping from data to visual properties, and that mapping determines what a reader can extract. Module 01 establishes that properly, because it reframes the whole subject: the question is not 'which chart looks good' but 'which channel carries this variable accurately enough for the judgement the reader has to make' — which has measured answers.

So the first third is the measurements. Some visual channels are read accurately and some are not, and the ranking is empirical (Modules 02 through 04) — which means a chart choice can be right or wrong rather than merely tasteful, and that is what makes this an engineering subject rather than a design one.

The second theme is that the hard cases are the ones with structure the plane does not have. Networks, hierarchies, high dimension, and geography all have to be projected into two dimensions (Modules 07 and 08), and every projection discards something — so the question becomes which distortion you are choosing and whether you said so.

The third is that uncertainty is almost always omitted and almost always matters. Module 09 is about showing it — because a point estimate drawn without its interval invites a conclusion the data does not support, and this is the commonest honest failure in the subject (the dishonest ones are Module 10).

And the closing position is about honest claims. Module 10 covers deliberate deception and the axis conventions, and Module 13 is about what a visualisation establishes — because 'the chart shows a clear trend' is a statement about the reader's perception as much as about the data.

Learning outcomes

On finishing this course you should be able to do the following without reference material.

- 1 Explain visualisation as an encoding and state what follows.
- 2 Rank the visual channels by accuracy and apply the ranking.
- 3 Match marks and channels to data types.
- 4 Use colour correctly and explain why most scales are wrong.
- 5 Choose a chart type from the judgement required.
- 6 Design multiple coordinated views and interaction.
- 7 Visualise networks and hierarchies, with their trade-offs.
- 8 Visualise spatial data and explain projection distortion.
- 9 Show uncertainty, and explain why it is usually omitted.
- 10 Recognise deceptive charts and state the axis conventions.
- 11 Visualise data too large to draw directly.
- 12 Explain scientific and volumetric visualisation.
- 13 State what a visualisation honestly establishes.

Module schedule

One module per week for thirteen weeks, then two weeks for the final project. Each module is a slide deck, a set of lecture notes, mapped external lectures, and exercises that must run.

Wk	Module	Core question it answers
1	Visualisation as Encoding	What is a chart, mechanically?
2	The Channel Ranking	Which visual channels are read accurately?
3	Data Types and Marks	Why can this variable not go on that channel?
4	Colour	Why is the rainbow scale a defect?
5	Chart Types	Which chart, and why that one?
6	Multiple Views and Interaction	What do you do when one chart is not enough?
7	Networks and Hierarchies	How do you draw a graph nobody can read?
8	Maps	Why is a map of counts a map of population?
9	Uncertainty	How sure is the line you drew?
10	Deception	How is this chart lying?

Wk	Module	Core question it answers
11	Scale	How do you draw a billion points?
12	Scientific Visualisation	How do you see inside a three-dimensional field?
13	Claiming a Visualisation Shows Something	The chart shows a clear trend. Does it?

Projects

Project 1 — One dataset, four encodings (after Module 6)

Encode the same data four ways and measure what each lets a reader extract.

Requirements

- **One dataset and four genuinely different encodings** of the same variables — not four styles of the same chart.
- **Each encoding justified against the channel ranking** (Module 02 §2), stating which variable got which channel and why.
- **Three specific questions a reader should be able to answer** from the data — a lookup, a comparison, and a trend.
- **Five readers tested** on those three questions, per encoding, with accuracy and time recorded.
- **A colour-vision check** and a greyscale check on all four.
- **And an account of which encoding won which question, and why that was predictable** from the ranking.

Done means

- **Four genuinely different encodings**, not four colour schemes — **which is graded first**, because the exercise is about channels.
- **The predictions made before the testing**, and the wrong ones reported (CSCE 671 Project 1's same requirement).
- **Accuracy and time**, because a slow correct reading is a different finding from a fast wrong one.
- **And the ranking used as a prediction rather than cited — the point is that it works**, and demonstrating that is the deliverable.

Project 2 — A real dataset, honestly shown (after Module 12)

Visualise something messy, including its uncertainty, and test what readers take away.

Requirements

- **A dataset with missing values, uncertainty, and more dimensions than fit** — and **all three handled explicitly rather than dropped**.
- **The uncertainty shown** (Module 09), not omitted — and the choice of representation justified.
- **A baseline**: the obvious chart somebody would have made, which yours has to beat on a stated question.
- **Five readers asked what the figure says**, in their own words, before being told.
- **Every misreading recorded**, with the encoding decision that caused it.
- **And an honest claim** in Module 13 §2's form, including what the figure does not show.

Done means

- **The uncertainty shown and the representation justified — a figure that omits it fails this project**, whatever else it does.
- **Readers asked open-endedly before being prompted**, because 'does this show X' gets a yes (CSCE 671 Module 09 §2).
- **The misreadings traced to encoding decisions**, which is the analytical skill being assessed.
- **And the not-shown list specific and non-empty. Every visualisation discards something**, and saying what is the difference between a figure and an argument.

Assessment

Nobody grades this. The standard is therefore mechanical, and you apply it to yourself honestly or the credential is worth nothing.

Component	Weight	Standard
Module exercises	40%	Code runs and produces the specified output. Not 'I understand how I would do it.'
Projects	45%	In the portfolio repository, with a README showing results and a written note on what was hard.
Self-checks	15%	Answered in writing from memory. Recognition is not knowledge; writing exposes the difference.

Required materials

All free. Nothing in this course requires a purchase.

Resource	Role
Munzner — Visualization Analysis and Design	The primary text. The what-why-how framework and the channel ranking this course uses come from here, and it is the most rigorous treatment available. Library copy; the slides are free from the author.
Ware — Information Visualization: Perception for Design, 4th edition	Modules 02 through 04. The perceptual research underlying every claim about which channels work, with the studies cited. Library copy.
Wilke — Fundamentals of Data Visualization (free online)	Modules 05, 09, and 10, free in full. Practical, opinionated, and unusually good on uncertainty and on the ways a chart misleads.
Tufte — The Visual Display of Quantitative Information	Module 10's conventions, and the data-ink argument. Influential, assertive, and worth reading critically — several of its claims did not survive measurement, which Module 05 §4 covers. Library copy.

Resource	Role
Healey & Enns — Attention and Visual Memory in Visualization (free)	Module 02's preattentive material , with interactive demonstrations that are worth running rather than reading.
The IEEE VIS proceedings (free preprints)	The field's primary literature . Module 13's exercises use it; the evaluation sections are where the honest claims are.

Tooling

- **D3 or Vega-Lite**, and **Vega-Lite first** — because **its grammar makes the encoding explicit**, which is exactly what Module 01 asks you to think about, and D3's flexibility hides it.
- **Python with matplotlib and altair**, or R with ggplot2 — **a grammar-of-graphics library, so the encoding is a stated mapping rather than a chart type**.
- **A colour tool that reports perceptual uniformity** — **Colorbrewer**, or the **viridis family** — and **Module 04 §2 explains why the default rainbow scale is a defect**.
- **A colour-vision simulator**, used on every figure before it ships (CSCE 632 §03).
- **A dataset you did not choose for its convenience**. **Project 2 requires one with missing values, uncertainty, and more dimensions than will fit** — which is what real data is like.
- **And three people who will tell you what they think your figure says**. **Module 13 §1's point is that you cannot read your own chart naively**, which is CSCE 671's argument in this subject.

Academic honesty, restated for a solo student

There is no one to cheat but yourself, which makes it easier, not harder. Two rules: copy code only after you have written a worse version yourself, and never let a working result stand in for an understood one. When you paste something you do not understand, write a TODO naming what you do not understand, and go back to it.